EXPERIMENT 1: Install a C Compiler in a Virtual Machine and Execute a Fibonacci Series Program Using C

1. Aim

To install a C compiler in a virtual machine and execute a C program to generate the Fibonacci series.

2. Procedure

1. Open virtualization software such as Oracle VirtualBox or VMware Workstation.
2. Start a virtual machine with an installed Linux operating system.
3. Open the Terminal application inside the Linux virtual machine.
4. Update the package repository using the command shown below.
5. Install the GCC compiler.
6. Verify the compiler installation by checking its version.
7. Create a C source file named fibonacci.c using a text editor.
8. Type the Fibonacci series program and save the file.
9. Compile the program using GCC to generate an executable file.
10. Run the executable, enter the number of terms and observe the series.

3. Program

Install and verify GCC:

```
sudo apt update
sudo apt install gcc -y
gcc --version
```

Program – fibonacci.c:

```
#include <stdio.h>

int main()
{
    int n, a = 0, b = 1, c, i;

    printf("Enter the number of terms: ");
    scanf("%d", &n);

    printf("Fibonacci Series:\n");

    for(i = 1; i <= n; i++)
    {
        printf("%d ", a);
        c = a + b;
        a = b;
        b = c;
    }

    return 0;
}
```

Compile and execute:

```
gcc fibonacci.c -o fibonacci
./fibonacci
```

4. Output of Program

```
Enter the number of terms: 10

Fibonacci Series:
0 1 1 2 3 5 8 13 21 34
```

5. Result

Thus, GCC was successfully installed in the virtual machine, and the C program to generate the Fibonacci series was compiled and executed successfully.

EXPERIMENT 1: Install a C Compiler in a Virtual Machine and Execute a Fibonacci Series Program Using C

1. Aim

To install and configure the GCC (GNU Compiler Collection) in a Linux-based virtual machine and develop, compile and execute a C program to generate the Fibonacci series for a given number of terms.

2. Procedure

1. Start the host computer and open virtualization software such as Oracle VirtualBox or VMware Workstation. 2. Start the Linux virtual machine and wait until the operating system is completely loaded. 3. Log in using the username and password configured during Linux installation. 4. Open the Terminal application to execute Linux commands. 5. Update the package repository using sudo apt update to obtain the latest available package information. 6. Install the GCC compiler using the command sudo apt install gcc -y. 7. Verify the installation by executing gcc --version. The installed compiler version will be displayed. 8. Create a new C source file named fibonacci.c using a text editor such as Nano or Gedit. 9. Enter the C program to generate the Fibonacci series and save the source file. 10. Open the terminal in the directory containing the source file. 11. Compile the program using gcc fibonacci.c -o fibonacci. This generates an executable file named fibonacci. 12. Execute the program using ./fibonacci and enter the required number of terms. 13. Observe the generated Fibonacci sequence and verify that every term is the sum of the previous two terms. 14. Repeat the execution with a different number of terms to confirm that the program produces the expected result.

4. Output of Program

Sample Input: Enter the number of terms: 10 Expected Output: Fibonacci Series: 0 1 1 2 3 5 8 13 21 34 Explanation: The Fibonacci series starts with 0 and 1. Every subsequent number is calculated by adding the previous two numbers. The program successfully generates the requested number of terms.

5. Result

Thus, the GCC compiler was successfully installed and configured in the Linux virtual machine. The C program for generating the Fibonacci series was written, compiled and executed successfully, and the expected sequence was obtained.